

Problem 1: Lay out cards 0–7, leftmost first. Cut into two equal rows without reversing either half. Interleave one card at a time, starting with the first half. The first shuffle is shown. Rebuild it with cards, then repeat the same rule until the original row first returns. Record every row.

Shuffle	Card labels, left to right
0	0 1 2 3 4 5 6 7
1	0 4 1 5 2 6 3 7
2	
3	
4	

Problem 2: Number places 0–7 from left to right on a strip of paper. Put the original row above it. Track card 1 through the shuffles, recording its place each time. Reset and track card 3, then card 7.

Problem 3: Reset to 0–7. Shuffle once. Fill where each old position moves. A card’s label stays the same; its position can change. For example, the card at old position 4 moves to position 1.

Old position	0	1	2	3	4	5	6	7
New position					1			

Problem 4: Draw arrow loops for this position map, starting at 0 and then at each unused position. Use the loops to predict the first return of the whole row. Check against page 1.

Problem 5: For positions 0–6, double the old position and subtract 7 if needed to leave a number from 0 to 6. Compare with your measured new positions. Treat position 7 separately.

Old position	0	1	2	3	4	5	6
Double							
Remainder after 7							

Problem 6: Starting with 1, repeatedly double and keep the remainder after division by 7. Record until 1 returns. Compare with the route of old position 1.

Problem 7: Use six cards 0–5. Repeat the same out-shuffle until the full row returns. Record rows. Then test whether the position rule is doubling with remainder after division by 5, with position 5 fixed.

Problem 8: For sixteen cards 0–15, predict the return by starting with 1, repeatedly doubling, and taking the remainder after division by 15. Check with cards or draw the position loops; keep position 15 fixed separately.

Problem 9: Compare the first return times for 6,8,16 cards. Record one comparison showing whether a larger deck must take more shuffles to return.